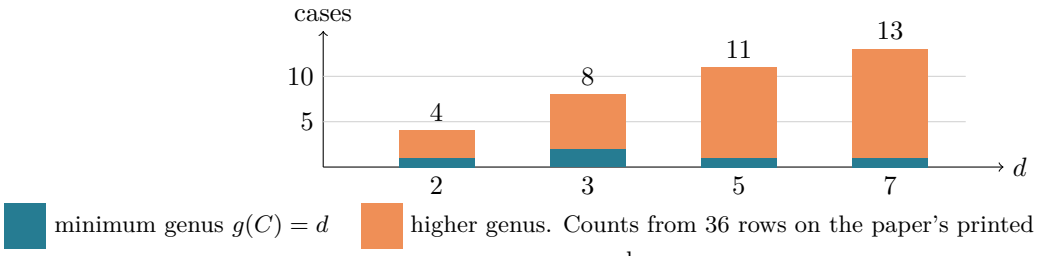
A Source-Verified Index of Odd-Characteristic Elliptic Covering Cases

Research note

12 October 2026

Contribution. The 36 numbered elliptic covering cases in Iijima–Momose–Chao's 2015 classification are now queryable by extension degree d, binary covering rank n, excess genus $e = g(C) - d$, and allowed coefficient square class. Five cases have the minimum genus $g(C) = d$; 31 have larger genus. This count is of published parameter families, not of exact elliptic models. The source is https://eprint.iacr.org/2015/805.pdf, Section 5.

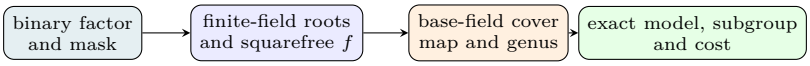


Counts from 36 rows on the paper's printed pages 17–18; one bar per degree.

d	cases	$g(C) = d$	higher genus	largest $g(C)$
2	4	1	3	5
3	8	2	6	17
5	11	1	10	129
7	13	1	12	769

The five minimum-genus cases are paper Cases 1, 5, 6, 13, and 24. The 2013 isogeny-condition list has 27 cases across elliptic and genus-two and genus-three source curves; it has a different scope and overlaps this table (https://eprint.iacr.org/2013/487.pdf).

Constructive screen. For $k = \mathbf{F}_q$ and $k_d = \mathbf{F}_{q^d}$, choose a listed binary factor $F \mid X^d + 1$ of degree n and set $\widehat{F} = (X^d + 1)/F$. The source's Procedure 1 chooses a coprime to F and uses $\Phi = a\widehat{F} \pmod{X^d + 1}$ as a binary Frobenius ramification mask. Cyclically rotated masks are equivalent. The native Rust replay records 27 mask orbits over the 11 listed factors. For the cubic $n = 2$ branch, $F = X^2 + X + 1$ and $\widehat{F} = X + 1$, giving mask 011 and roots $\{\alpha, \alpha^q\}$. Case 5's source polynomial has four factors $(x - \alpha_1)(x - \alpha_1^q)(x - \alpha_2)(x - \alpha_2^q)$; the separate cubic $n = 3$ minimum-genus branch is Case 6.



Current native and Lean checks address the first gate; an exact curve requires certificates for the later gates.

To realize a case, include every conjugate root over k_d, choose a squarefree cubic or quartic f, enforce the paper's condition $F(\sigma)f = 1$ in $k_d(x)^*/k_d(x)^{*2}$, then verify the base-field model, covering map, genus, exact twist, and selected-subgroup transport. A binary mask by itself is a candidate. The two factorization identities for degrees three and seven are Lean-checked with only standard Lean foundations; formalizing the covering classification remains open. The accompanying `REPORT.md`, `PROOF_MAP.md`, and SQLite file record the source, formalization status, and reproducible queries.

A checked negative discriminator. Costa et al., Theorem 2.8 (https://arxiv.org/pdf/2002.02067), rules out a genus-three hyperelliptic Jacobian when its second Weil coefficient is even and its third is odd. A cubic restriction with polynomial $T^6 - tT^3 + q^3$ meets that excluded pattern for odd t. Direct database queries found zero odd traces in 87,486 complete exact ordinary rows and 246,512 ordinary cubic diagnostic rows. Thus this parity test separates none of the unresolved rows in the current corpus; it addresses only the hyperelliptic branch.